Habitat-dependent consequences of including lagged responses to climate in structured population models

OR

Including lagged responses to climate in structured population models has habitat-specific implications

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Abstract

1. There is increasing evidence that changes in demographic vital rates due to contemporary climatic conditions can be delayed months or even years. Moreover, the temporal delay between an climatic event and the resulting changes in vital rates may depend on both the intensity of the event and its timing relative to the physiological mechanisms underlying the response.
2. Because lagged effects are often directly linked to reproduction and survival, it has been argued they could have major but underappreciated consequences for population dynamics. There have been few tests of this hypothesis, however, in part because the long-term data required to incorporate lagged effects in structured population models are rarely available.
3. We parameterized three classes of Integral Projection Models — deterministic, stochastic, and stochastic with lagged effects of precipitation on vital rates — using a decade of annual censuses from populations of an Amazonian understory herb. We then evaluated whether (1) if incorporating lagged effects altered projections of population growth rate, and (2) if habitat-specific differences in lagged effects translate into habitat-specific projections of population growth rate (i.e., λ). 4. Projections of λ were lower when models included lagged effects; projections of population structure were also more variable both within and across habitats.
4. Synthesis: These differences could be critical in conservation or management settings, where even small differences in λ can have major consequences for population dynamics.

Keywords: demography, environmental stochasticity, integral projection models, lagged effects, structured population models, population dynamics

Manuscript elements: Figures 1-4, Tables 1-6

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Introduction

Current climatic conditions can have immediate impacts on the growth, survival, or reproduction of plants (Evers et al., 2023; Williams et al., 2025). However, there is increasing evidence for potential delays of months or even years between current conditions and any resulting changes in demographic vital rates (Evers et al., 2021; Prather et al., 2023; Scott et al., 2021; Tenhumberg et al., 2018). These *Lagged Effects*, also known as *Delayed Life-history Events* (Beckerman et al., 2002) or *Temporally Varied Responses* (Evers et al., 2023), can simultaneously affect an entire cohort (e.g., all seedlings emerging during a drought have delayed maturation) or only a subset of the population (e.g., a freeze reduces flowering by small trees but not large ones). Moreover, the delay between an environmental event and altered vital rates can depend on both the intensity of the event and its timing relative to the underlying physiological mechanisms (Criley & Lekawatana, 1994; Evers et al., 2021). For example, a drought during the early phase of floral bud formation might have a much larger impact on the number of flowers of fruits a plant produces than one at a later stage of development (sensu Tenhumberg et al., 2018).

Because many Lagged Effects appear to influence plant reproduction and survival (e.g., Prather et al., 2023), it has been argued they could have major consequences for population dynamics (Beckerman et al., 2002; Tenhumberg et al., 2018). Despite this, very few studies have explicitly tested for lagged effects of climate on vital rates (reviewed in Evers et al., 2021) or included lagged effects in demographic models (*sensu* Tenhumberg et al., 2018; Kiss et al., 2024; Mensah et al., 2025). In perhaps the most comprehensive study to do so, Evers et al. (2023) found that the length of the lag (i.e., the "temporal window") influenced the effect of climate variability on long-term population growth rates (i.e., λ; see also Tenhumberg et al. (2018)). However, it remains unknown whether models with and without lagged effects actually project different values of λ or how the effects of temporal lags compare with those of other demographic drivers. For example, it has also been suggested that the demographic consequences of lagged effects could vary strongly both among and between populations (Scott et al., 2021). If individuals in some locations or (micro)habitats are buffered against climatic extremes (Williams et al., 2025), this could alter the magnitude of delayed effects or which vital rates they affect (Scott et al., 2021). A general understanding of how climate shapes plant demography requires understanding not only when and how to incorporate lagged effects in models (Evers et al., 2023), but also how the demographic impacts of climate compare to those of landscape structure, disturbance, biotic interactions, or other drivers of population dynamics (Williams et al., 2025).

Efforts to include lagged effects in structured population models are rare in part because experimental tests of these effects are challenging to design, implement, and maintain (Kuss et al., 2008). While observational data can also be used to detect lagged effects, doing so requires long-term data on both the potential lagged effects and their putative climate drivers. Not only are such coupled data sets extremely rare (*sensu* Evers et al., 2021), the methods for both identifying lagged effects (e.g., Ogle et al., 2014; Evers et al., 2021; Scott et al., 2021) and modeling their demographic impacts (Evers et al., 2023; Mensah et al., 2025) can be challenging to implement. For example, many of the potentially applicable statistical methods have stringent assumptions and data requirements rarely met by ecological data (Evers et al., 2023; Metcalf et al., 2015). Terms for complex biological processes such as lagged effects can also result in demographic models for which mathematical solutions are intractable or make analysis via simulation more difficult. Addressing these challenges is a major undertaking, the value of which depends on the effort required to consider lagged effects with more complex models vs. the potential consequences of failing to do so — consequences that could include overestimating projections of population growth rate (i.e., λ) in a conservation setting to drawing invalid conclusions regarding support for the predictions of ecological or evolutionary theory.

Integral Projection Models (i.e., IPMs) are an important and widely used tool for studying demography and population dynamics (Ellner & Rees, 2006; Rees et al., 2014; Rees & Ellner, 2009). Their flexibility, in concert with a rapidly growing suite of software, data, and other resources (Ellner et al., 2016; Levin et al., 2021; Salguero-Gómez et al., 2015), have facilitated their use to study a wide range of topics in ecology, evolution, and conservation (Crone et al., 2011; Ellner et al., 2016; Morris & Doak, 2002). Mathematical and statistical advances (e.g., Williams et al., 2012; Brooks et al., 2019) have rapidly expanded the scope of questions and biological processes that can be investigated with these models (e.g., Metcalf et al., 2015; Ellner et al., 2016; Rees & Ellner, 2016). Here we used three classes of Integral Projection Models — a deterministic IPM, a stochastic IPM, and a stochastic IPM with lagged effects of precipitation on vital rates — to address the following questions: First, how similar are the projections of λ from stochastic IPMs with and without lagged effects? Second, do habitat-specific differences in lagged effects of precipitation on vital rates translate into habitat-specific projections of λ? The models were parameterized using long-term data collected on populations of an Amazonian understory herb (*Heliconia acuminata*, Heliconiaceae) found in either Continuous Forest or Forest Fragments; the lagged effects of precipitation extremes on *H. acuminata* growth, reproduction, and survival can be delayed up to 36 months, with the duration and magnitude of these lagged effects varying by vital rate and habitat (Scott et al., 2021).

Materials and Methods

Study System and Demographic Data

Heliconia acuminata (Heliconiaceae) is a perennial, self-incompatible monocot that is distributed throughout much of the Amazon basin (J. Kress, 1990). While some *Heliconia* species grow in large aggregations on roadsides, gaps, and in other disturbed habitats, others, including *H. acuminata*, grow primarily in the forest understory (W. J. Kress, 1983; Ribeiro et al., 2010). Understory *Heliconia* species produce fewer flowers and are pollinated by traplining hummingbirds (Bruna et al., 2004; Stouffer & Bierregaard, 1996). The models and analyses here are based on 11 years (1998-2009) of demographic data collected on >8500 *H. acuminata* found at Brazil's Biological Dynamics of Forest Fragments Project (BDFFP), located ~70 km north of Manaus, Brazil. The BDFFP reserves include both continuous forest and forest fragments that range in size from 1-100 ha. These fragment reserves were originally isolated in the early 1980's by the creation of cattle pastures, with the secondary growth surrounding them periodically cleared to ensure their continued isolation. The habitat in all sites is non-flooded lowland rain forest with rugged topography. A complete summary of the BDFFP and its history can be found in Bierregaard et al. (2001).

A complete description of our demographic methods, data, and analyses to date can be found in Bruna et al. (2023). Briefly, in 1997–1998 a series of 5000 m^2 plots were established in the BDFFP reserves: N=6 in Continuous Forest and N=4 in 1-ha Forest Fragments (i.e., CF and FF, respectively). All of the *Heliconia acuminata* in these plots were marked and measured; the plots were censused annually, at which time a team recorded the size of surviving individuals, marked and measured new seedlings, and identified any previously marked plants that died. Each plot was also surveyed 4-5 times during the flowering season to identify reproductive plants. These surveys were complemented by data on the number of fruits per flowering plant (Bruna, 2021) and seeds per fruit (Bruna, 2014) that were collected outside of the demographic plots to avoid altering within-plot recruitment. We also conducted experiments to quantify the probability of seed germination and seedling establishment in both forest fragments and continuous forest (Bruna, 1999; Bruna, 2002).

The rainy season in our sites is typically from late December through late May. *Heliconia acuminata* in our site begin flowering early in the rainy season and most reproductive plants produce a single infloresence (range = 1–7) with 20–25 flowers (Bruna & Kress, 2002). Fruits mature April-May and have 1–3 seeds per fruit ($\bar{x} = 2$) that are dispersed by a thrush and several species of manakin (Uriarte et al., 2011). Dispersed seeds germinate approximately 6 months after dispersal at the onset of the subsequent rainy season, with rates of germination and seedling establishment higher in continuous forest than forest fragments (Bruna, 1999; Bruna, 2002). The median number of plants in CF plots was over twice that of the plots in 1-ha fragments (CF: median = 788, range = 201-1549; 1-ha: median = 339, range = 297-400, Bruna et al. (2023)).

Construction of Integral Projection Models

We projected the growth rate and structure of *Heliconia acuminata* with three classes of IPMs: (1) deterministic, (2) stochastic, and (3) stochastic with lagged environmental effects. Each of these IPMs required different functional forms of the underlying vital rate functions used to describe the *H. acuminata* life cycle (Figure 1). All models were density-independent, with the deterministic model serving as the foundation for the more complex models. Our modeling workflows were managed using the `targets` R package (Landau, 2021); in addition to ensuring reproducibility this allowed us to track computational time spent processing and analyzing each class of IPM.

(1) Deterministic IPM: In this model the size and structure of a population in year $t+1$ is determined by the survival and change in size of the n plants of size z that are alive in year t, given by

$$n(z', t+1) = \int_L^U P(z', z)n(z, t)\ \mathrm{d}z + R(z')n_s(t) \tag{1}$$

plus the number of new seedlings that entered the population, given by

$$n_s(t+1) = \int_L^U F(z)n(z, t)\ \mathrm{d}z \tag{2}$$

Equation 1 has two components. The first is sub-kernel $P(z', z)$, which describes the size-dependent probability of survival of mature (i.e., post-seedling) plants, s, and their growth (or regression) from size z to z':

$$P(z', z) = s(z)G(z', z) \tag{3}$$

The second component is sub-kernel $R(z')$. This sub-kernel describes the survival of seedlings to year $t+1$ of seedlings that established in year t and the size of these surviving seedlings when they transition into the 'mature' plant population after surviving one year:

$$R(z') = s_s G_s(z') \tag{4}$$

Note that in Equation 4 both the probability that newly established seedlings survive their first year, s_s, and the size to which they grow at the end of this year, G_s, are size-independent. IPMs can readily accommodate such transitions of individuals from a discrete state to a continuous one (i.e., from 'seedling' to 'mature plant of size z',' Ellner et al., 2016); we treated seedlings as a distinct and discrete category in our models because they have lower survival and growth in their first year than comparably sized but plants that have been established for over one year (Bruna, 2003; Scott et al., 2021).

The number of new seedlings entering the population in year $t+1$ is a function of the number of 'mature' plants in year t and a sub-kernel describing the fecundity of these individuals:

$$F(z) = p_f(z)f(z)g \tag{5}$$

In this sub-kernel both the probability that a mature plant will flower, p_f, and the number of seeds a flowering plant will produce, f, are size-dependent, but all seeds germinate and establish as seedlings with probability g.

We used the annual census data (Bruna, 2003) to fit the deterministic vital rate functions for growth, survival, and flowering in each habitat type (Table 1). The data from all plots within a habitat class were pooled to create a single 'summary population' representing that habitat type; the summary populations were then used to fit the vital rate functions. A summary population is a superior means of synthesizing the demography of multiple populations than calculating an average vital rate for multiple populations because it corrects for the disproportionate weight that the low plant numbers in some size classes in some locations can give to vital rates [; Bruna (2003)]. For established plants the vital rates were modeled as a smooth function of size in the previous census with generalized additive models (GAMs) fit with the `mgcv` package (Wood, 2011) for the R statistical programming language (R Core Team, 2020). For consistency, seedling survival and growth were also modeled using GAMs, but without size in the previous census as a predictor (i.e. 'intercept-only' models). For growth models a scaled t family distribution provided a better fit to the data than a Gaussian fit, as the residuals with a simple Gaussian model were leptokurtic. To model size-specific fecundity we used data from prior studies on the number of fruits per flowering plant (Bruna, 2021), seeds per fruit (Bruna, 2014), and experimentally derived estimates of seed germination and seedling establishment (Bruna, 1999; Bruna, 2002).

(2) Stochastic IPMs: To include temporal stochasticity in our IPMs we included a random effect of year. This was done using a factor–smooth interaction that allowed the functional form of the relationship between plant size and vital rates to vary among transition years (Figure 1). We generated kernels for every transition year using the long-term survey data (Bruna et al., 2023) and random smooths for year. We then randomly selected one of these sets of kernels to use in each iteration of the IPM. This procedure is equivalent to *'kernel resampling'* (*sensu* Metcalf et al., 2015) or matrix selection for matrix population models (Boyce et al., 2006; Caswell, 2001).

(3) Stochastic IPMs with lagged effects of precipitation on vital rates: We explicitly modeled the lagged effects of precipitation extremes on vital rates using the procedure described in Scott et al. (2021).

Briefly, we first calculated the Standardized Precipitation Evapotranspiraton Index (i.e., SPEI) for our study site using a published gridded dataset based on ground measurements (Xavier et al., 2016). After we fit vital rate models using the long-term survey data (Figure 1), we modeled delayed effects of SPEI with Distributed Lag Non-linear Models (i.e., DLNMs) with a maximum lag of 36 months (Scott et al., 2021). To iterate these parameter-resampled IPMs (*sensu* Metcalf et al., 2015) we first created a random sequence of SPEI values by sampling years of the observed (monthly) SPEI data. For every year we then calculated a lag of 36 months from the month in which that year's census was completed. These values were then used to predict the fitted values from vital rate models, which generated different kernels for each iteration of the IPM. The kernels of successive iterations are not entirely independent — the SPEI values used to calculate vital rates include values used in the previous two iterations — but they are ergodic.

All IPMs were constructed and iterated using the `ipmr` package (Levin et al., 2021) for the R statistical programming language (R Core Team, 2020). The IPMs used 100 meshpoints and the midpoint rule for calculating kernels (Rees et al., 2014). For each type of IPM we iterated the model for 1000 time steps, but discarded the first 100 time steps to omit transient effects. Stochastic growth rates (λ_s) were calculated as the average $ln(\lambda)$ from each time step (Caswell, 2001) and then back-transformed to allow for direct comparison with projections of λ from deterministic models. The initial starting vector primarily influences a population's transient dynamics; we therefore used the distribution of established plant sizes and proportion of seedlings in the full demographic data set as the initial population vector for all models.

Finally, we estimated the 95% confidence intervals for each IPMs projections of λ. To do so we first created 500 populations for each habitat type by sampling individual plants with replacement (i.e., bootstrapping) until the population size of each matched that of the initial population vector. We then re-fit the vital rate models for growth, survival, and flowering for each of these bootstrapped population and constructed new IPMs for each population as above (the models for germination and establishment rate, fruits per flowering plant, and seeds per fruit were not refit because these were estimated using different data sets). The projections of λ for the new populations were then used to estimate the upper and lower 95% bias-corrected percentile intervals (Caswell, 2001; Manly & Manly, 2018).

Statistical analyses

Comparison of λ in Continuous Forest and Forest Fragments: To determine if the deterministic projections of λ for the populations in Continuous Forest (i.e, λ^{CF}) and Forest Fragments (i.e., λ^{FF}) were significantly different we used the randomization test procedure described by Caswell (2001). Briefly, we randomly assigned plants with their demographic history among two populations, R_1 and R_2, that were equal in size to the original CF and FF populations. We then calculated λ^{R_1}, λ^{R_2}, and the absolute value of the difference between the two (i.e., θ). This was repeated N = 1000 times, after which we determined the proportion of simulations in which θ was greater than the observed difference between λ^{CF} and λ^{FF} (i.e., $P[\theta \geq \theta_{obs}]$).

We used a Generalized Linear Model to compare the projections of λ from the stochastic IPMs with and without lagged effects. We modeled λ_i as a function of IPM type (i.e., with and without lagged effects) and habitat (i.e., continuous forest vs. fragments), and the interaction of IPM type and habitat. Because the response variable (λ) was continuous we used the Gaussian distribution with the identity link function in our model. We verified the model assumptions by plotting residuals versus fitted values.

Population Structure: Finally, we compared the structure of populations that were projected through the first 250 time steps in each habitat by the IPMs with and without lagged effects. At each time step, we assigned the individuals to one of four stage classes based on plant size, probability of survival, and probability of reproduction (see Table 2). We then calculated the proportion of the population that was in each stage class in each habitat for each IPM type. Comparing the means and variances of these proportions will allow us to determine (1) if the two IPMs project different population structures over time, (2) if the population structure for a given IPM is consistent across habitat type. Because Shapiro-Wilk tests indicated that the distributions of proportions were not normally distributed, we used the non-parametric Ansari-Bradley Test to compare the distribution of proportions in each habitat x stage combination. These analyses were conducted using the R packages `rstatix` and `vartest`, respectively (Cosar & Dag, 2024; Kassambara, 2023).

Results

All IPMs projected higher population growth rates for *Heliconia acuminata* populations in Continuous Forest than for those in Forest Fragments (Table 3). The differences between λ^{CF} and λ^{FF}, which ranged from 1.19-2%, were significant for both deterministic ($P[\theta \geq \theta_{obs}] = 0$, N = 1000 randomization) and stochastic IPMs; there were also significant effects of IPM type, habitat, and their interaction on λ (Table 4). Because stochasticity is predicted to reduce population growth rates (Doak et al., 2005; Metcalf et al., 2015; Tuljapurkar et al., 2003), we were surprised to find that deterministic λ was nearly identical to the average λ from standard Stochastic IPMs (i.e., without lagged effects, Figure 2). However, we did see reductions in population growth rate when including lagged effects in IPMs: on average the projections of λ from these models were 1.5-2% lower, with 70-76% of the values below deterministic λ (Figure 2). The IPMs with lagged effects also appear to be more accurate — the underlying vital rate models for IPMs with lagged effects all had the best fit to the survey data (dAIC = 0, Table 1).

Including lagged effects in IPMs also resulted in significantly less predictable projections of population structure, both within (Figure 3A) and across habitats (Figure 3B, Table 5). This variability is particularly notable in forest fragments (Figure 4, Table 6), where we have previously shown lagged effects have very large impacts on growth and survival (Scott et al., 2021). The deterministic IPM for Continuous Forest projected slightly more of the smallest and largest plants than the one for Forest Fragments. In other words, populations in forest fragments had proportionately less recruitment and fewer individuals growing into the larger, reproductive size classes (Bruna & Oli, 2005). This shift towards intermediately sized, pre-reproductive plants in forest fragments is even more dramatic when using the stochastic IPM with lagged effects (Figure 3B).

Discussion

What are the implications of these results for the use of IPMs to study plant demography? The first is that failing to include lagged effects in models could significantly underestimate the growth rate of populations. While relative differences in projections of λ may be sufficient for some tests of theory, the actual value of projections may be critical in conservation or management settings. For example, they could be used to identify which populations face the greatest risk of extinction (e.g., $\lambda = 0.99$ vs. $\lambda = 1.01$) or, because even small differences in λ can have major short-term consequences for population dynamics (e.g., $\lambda = 1.01$ vs. $\lambda = 1.02$), be used to choose among alternative management strategies. Given the potential for lagged effects to either buffer or amplify the effects of climatic variability (Evers et al., 2023), including lagged climatic effects in models could also be especially critical for forecasting how plant populations will respond to climate change (Evers et al., 2023).

The second implication is that demographic precision is computationally costly. Our analyses were conducted on the University of Florida's HiperGator supercomputer using one nVidia A100 GPU with an allocation of 8 GB RAM. While each iteration of the Deterministic IPM and Stochastic IPM without lagged effects took only ~0.02 and ~0.07 min to iterate (respectively), an iteration of the Stochastic IPM with lagged effects took ~87.12 min. This is largely due to the required computational resources and algorithms (i.e., `predict()`), which are much slower for Generalized Additive Models with 2D smooths than for the GAMs used in Deterministic and Stochastic IPMs. Advances in computational power and access to supercomputer could lower the temporal and computational cost of running the thousands of iterations required to estimate λ and its confidence intervals, as could GAMs for large datasets (i.e., Wood, 2011). Until these resources are more broadly available, however, it will be important to consider if any model assumptions can be relaxed to reduce the "computational burden" (*sensu* Evers et al., 2023) of including lagged effects or if failing to include lagged effects in demographic models will alter the conclusions drawn from their analysis. Indeed, Compagnoni et al. (2024) found that using highly sophisticated antecedent effect models (Ogle et al., 2014) instead of simpler lieaner models did not significantly improve the ability to predict the effects of climate on vital rates. They also noted, however, that these results may have been due to climate signals that were too weak or noisy to influence the demography of their primarily temperate focal species.

It is important to emphasize, however, that computational power cannot compensate for limited data. Detecting lagged effects of climate and evaluating their consequences requires long-term demographic data — data that are only available for a relatively small number of species, of which few are in the tropics (Bruna et al., 2009; Römer et al., 2024) and where the synergistic effects of climate- and land-use change may be particularly acute (Laurance et al., 2014; Opdam & Wascher, 2004; Selwood et al., 2015). The increasing evidence that lagged effects are ubiquitous, and that they can have major demographic impacts, underscores the need to support the collection of such long-term data, the complementary development of experimental and statistical approaches to disentangling lagged effects, and community driven efforts to identify priority or model systems for in-depth investigation.

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CRediT Statement

The authors made the following contributions: Eric R. Scott: conceptualization, methodology, formal analysis, validation, visualization, data curation, software, writing – original draft, review, and editing; Emilio M. Bruna: conceptualization, methodology, formal analysis, validation, visualization, data curation, software, writing (original draft, review and editing), funding acquisition, project administration, supervision; Maria Uriarte: conceptualization, methodology, writing (review and editing), funding acquisition.

Data Availability Statement

Data and R code used in this study are archived with Zenodo at *(doi and url to be added on acceptance).*

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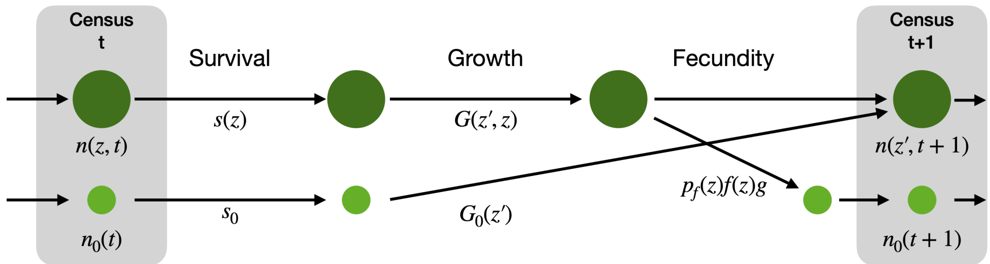

Description	Deterministic	Stochastic, kernel-resampled	Stochastic, parameter-resampled
Survival	$s(z)$	$s_y(z)$	$s(z; \theta_{0-36})$
Growth	$G(z'; z)$	$G_y(z'; z)$	$G(z', z; \theta_{0-36})$
Flowering	$p_f(z)$	$p_{f_y}(z)$	$p_f(z; \theta_{0-36})$
Size-specific fecundity	$f(z)$	$f(z)$	$f(z)$
Germination & establishment	g	g	g
Seedling survival	s_0	s_{0_y}	$s_0(\theta_{0-36})$
Seedling growth	$G_0(z')$	$G_{0_y}(z')$	$G_0(z'; \theta_{0-36})$

Figure 1: Life cycle diagram of *Heliconia acuminata*. Each transition is associated with an equation for a vital rate function. The functions shown on the diagram correspond to those used to construct a general, density-independent, deterministic IPM. The table below shows the equivalent equations for stochastic IPMs with and without lagged effects (i.e., kernel-resampled vs. parameter-resampled IPMs).

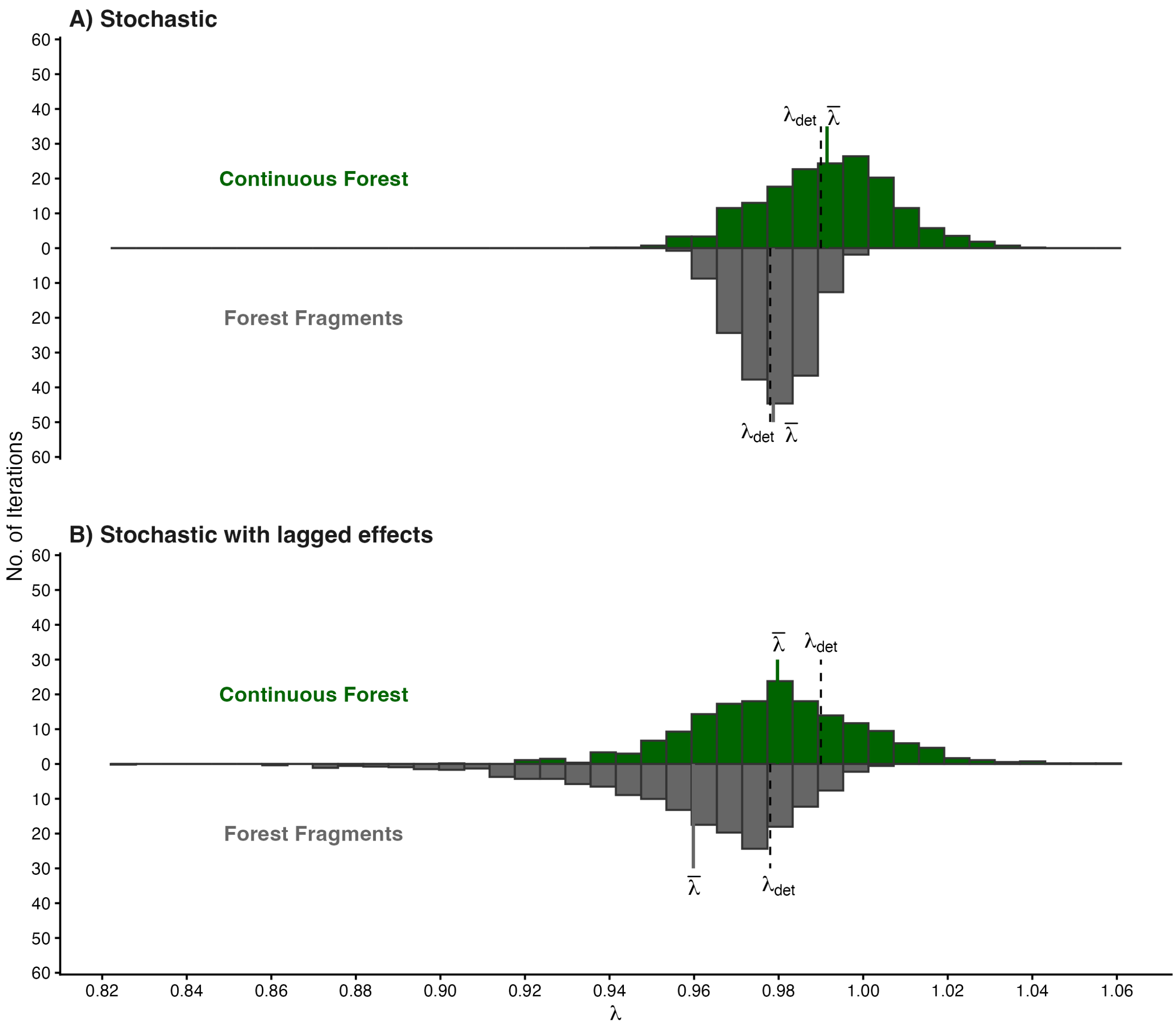

Figure 2: Distribution of 900 values of λ projected with (A) Stochastic IPMs without lagged effects and (B) Stochastic IPMs with lagged effects. IPMs were used to project λ for both Continuous Forest (above, in green) and Forest Fragments (below, in gray). The solid line indicates the mean value of λ, the dashed line indicates the value of λ in that habitat projected with Deterministic IPMs.

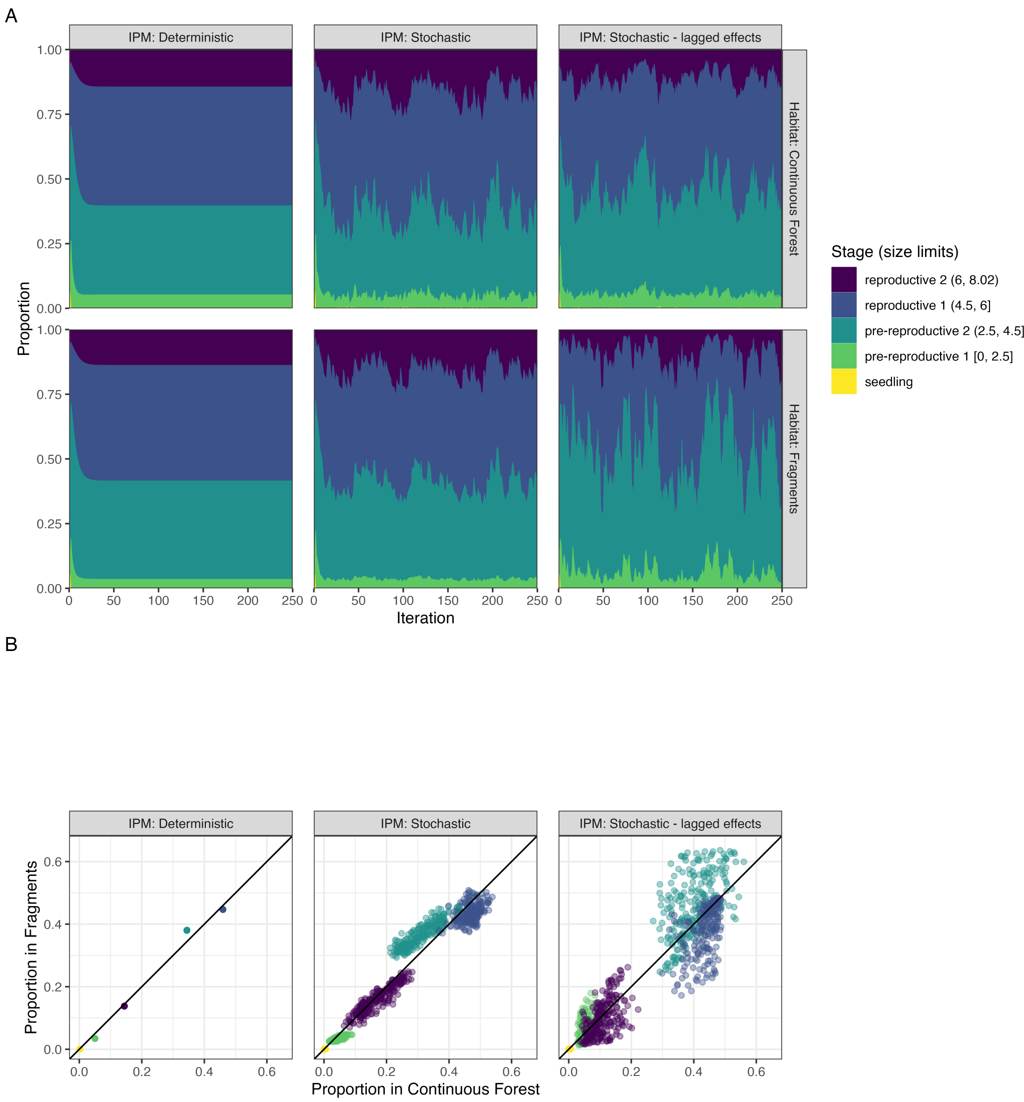

Figure 3: **(A)** The change over time in the the proportion of *Heliconia acuminata* populations in different size/stage classes when simulating population dyammics in Continuous Forest (CF) or Forest Fragment (FF) with three different integral projection models (e.g., IPMs). Results are shown for the first 250 iterations of populations; for the criteria used to define the size categories see Table 2. **(B)** The relative proportion of the population in each size class (FF vs. CF) for 250 iterations of each IPM model. Note that this excludes transient dynamics (iterations 1-30). Values on the 1-1 line indicate an iteration where CF and FF have the same proportion of the population in a given size class.

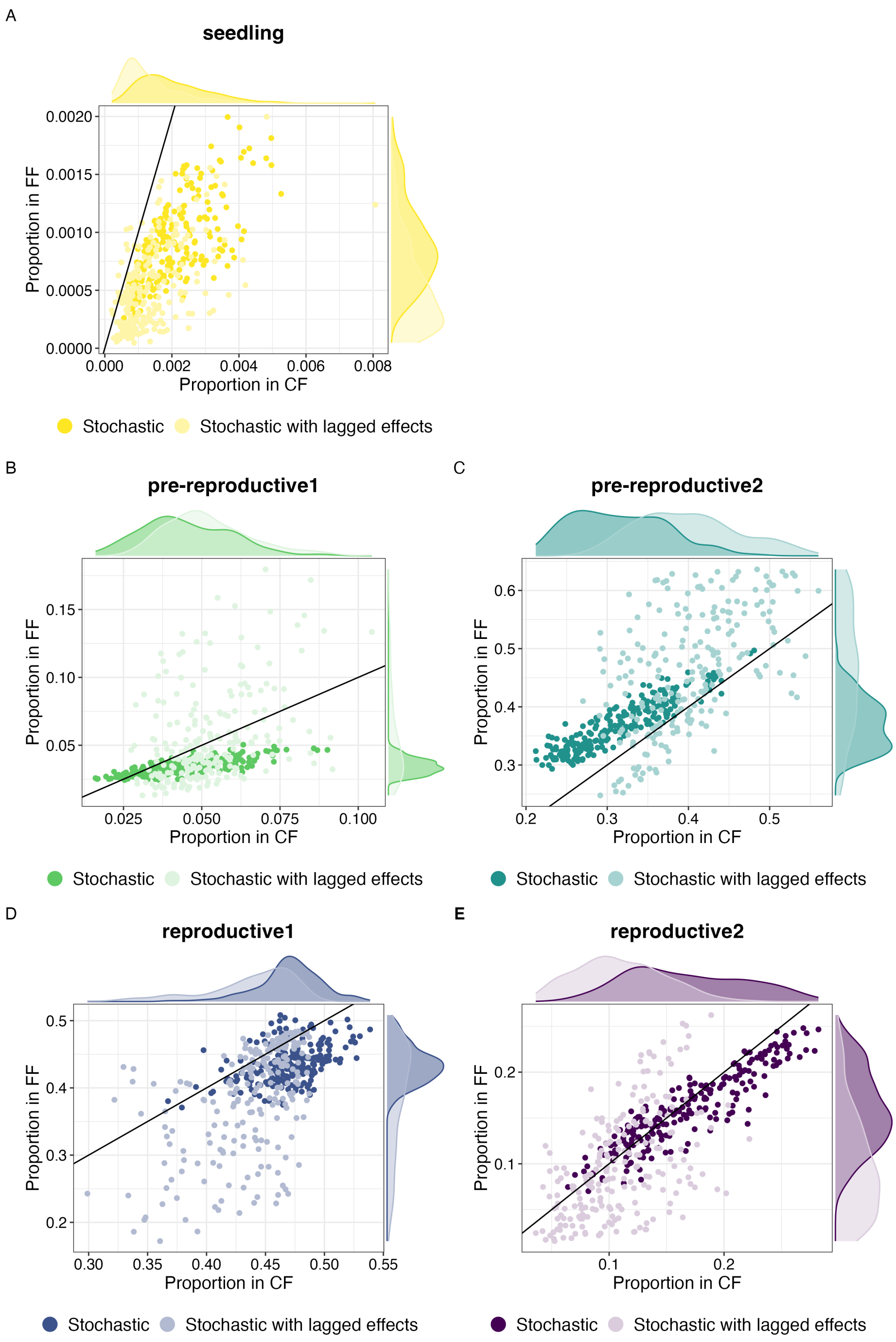

Figure 4: The relative proportion of the population in each size class (FF vs. CF) for each of 250 iterations of the no-lag IPMs (dark shading) and Stochastic IPMs with lagged effects (light shading). Values on the 1-1 line indicate an iteration where CF and FF have the same proportion of the population in a given size class; the marginal plots indicate the distribution of these relative proportions for each class of IPM in each habitat. Note that both the scatterplots and marginal plots exclude transient dynamics (iterations 1-30).

Table 1: Comparison of vital rate models used to build IPM. The 'Effect of Environment' column describes how environmental effects were included in models. Those with 'none' were used to build deterministic IPMs; those with a random effect of year were used to build stochastic IPMs without lagged effects (i.e., kernel-resampled models); and those with a distributed lag non-linear model (DLNM) were used to build stochastic IPMs with lagged effects (i.e., parameter-resampled models). 'edf' is the estimated degrees of freedom of the penalized GAM. ΔAIC is calculated within each habitat and vital rate combination. ΔAIC within 2 indicates models are equivalent.

Habitat	Vital Rate	Effect of Environment	*edf*	ΔAIC
Continuous Forest				
	Survival	Random effect of year	43.26	0.00
		DLNM	19.72	78.92
		None	4.98	260.01
	Growth	Random effect of year	78.43	0.00
		DLNM	23.87	158.46
		None	7.81	1896.03
	Flowering	DLNM	19.59	0.00
		Random effect of year	17.19	1.63
		None	7.47	381.86
	Seedling survival	None	1.00	0.00
		Random effect of year	1.82	1.39
		DLNM	4.01	1.53
	Seedling growth	Random effect of year	9.47	0.00
		DLNM	8.95	2.90
		None	1.00	172.33
Forest Fragments				
	Survival	DLNM	14.95	0.00
		Random effect of year	19.21	35.68
		None	4.33	51.25
	Growth	DLNM	25.18	0.00
		Random effect of year	37.84	199.98
		None	5.60	382.76
	Flowering	DLNM	20.61	0.00
		Random effect of year	13.81	27.40
		None	5.01	101.70
	Seedling survival	DLNM	5.57	0.00
		Random effect of year	5.09	5.72
		None	1.00	6.49
	Seedling growth	Random effect of year	6.25	0.00
		DLNM	8.18	2.29
		None	1.00	5.74

Table 2: Size and stage categories used for comparing *Heliconia acuminata* population structure. Note that seedlings are a discrete size class not based on size (see *Methods* for additional details).

Category	Log(size)	Avg. prob. survival	Prob. flowering
Seedlings	-	-	-
Pre-reproductive 1	0–2.5	≤ 0.9	≈ 0
Pre-reproductive 2	2.5-4.5	≥ 0.8	≈ 0
Reproductive 1	4.5-6	≥ 0.95	≤ 0.25
Reproductive 2	≥ 6	≥ 0.95	≥ 0.2

Table 3: Population growth rates for continuous forest (CF) and forest fragments (FF) under different kinds of IPMs with bootstrapped, bias-corrected, 95% confidence intervals.

IPM	Habitat	λ	95% CI (Lower, Upper)
Deterministic			
	FF	0.9778	(0.9736, 0.9823)
	CF	0.9897	(0.9877, 0.9920)
Stochastic			
	FF	0.9787	(0.9735, 0.9835)
	CF	0.9913	(0.9892, 0.9939)
Stochastic with lagged effects			
	FF	0.9595	(0.9459, 0.9689)
	CF	0.9795	(0.9752, 0.9867)

Table 4: Estimated parameters, standard errors, t-values and P-values for the GLM of the effect of Habitat and IPM Type on projections of lambda.

Term	Estimate	SE	z value	P
(Intercept)	0.98	0	1568.20	< 0.0001
$Habitat_{FF}$	-0.02	0	-22.52	< 0.0001
IPM_{Stoch}	0.01	0	13.27	< 0.0001
$Habitat_{FF} : IPM_{Stoch}$	0.01	0	5.76	< 0.0001

Table 5: Results of statistical tests comparing the variance in the proportion of each habitat's population projected to be in each stage class by Stochastic IPMs with and without lagged effects (N = 220 projections per stage class). The variances for each habitat x stage class x IPM combination can be found in Table 6. Comparisons where the p-value of the test was < 0.05 are indicated with an asterisk.

Stage	Habitat	Statistic	*df*	*P*
Seedling				
	CF	4.59	1	0.032*
	FF	11.51	1	0.001*
Pre-reproductive 1				
	CF	8.73	1	0.003*
	FF	96.13	1	< 0.0001*
Pre-reproductive 2				
	CF	0.99	1	0.319
	FF	39.67	1	< 0.0001*
Reproductive 1				
	CF	0.57	1	0.452
	FF	61.58	1	< 0.0001*
Reproductive 2				
	CF	0.11	1	0.745
	FF	18.78	1	< 0.0001*

Table 6: Summary statistics (median, mean, and variance) describing the proportion of populations projected to be in each of five life-history stages by Stochastic IPMs with and without lagged effects (i.e., kernel- vs. parameter-resampled IPMs, N = 220 projections for each IPM class x habitat combination.)

		Median		Mean		Variance	
Stage	IPM	CF	FF	CF	FF	CF	FF
Seedling							
	Stochastic	0.0018	0.0009	0.0020	0.0020	0.000001	0.000000
	Stochastic with lagged effects	0.0011	0.0004	0.0014	0.0014	0.000001	0.000000
Pre-reproductive 1							
	Stochastic	0.0426	0.0340	0.0450	0.0450	0.000203	0.000034
	Stochastic with lagged effects	0.0499	0.0438	0.0522	0.0522	0.000192	0.001167
Pre-reproductive 2							
	Stochastic	0.3097	0.3677	0.3147	0.3147	0.003354	0.001884
	Stochastic with lagged effects	0.3975	0.4613	0.4002	0.4002	0.003934	0.010380
Reproductive 1							
	Stochastic	0.4725	0.4332	0.4704	0.4704	0.000736	0.000822
	Stochastic with lagged effects	0.4464	0.4049	0.4371	0.4371	0.001481	0.006388
Reproductive 2							
	Stochastic	0.1630	0.1554	0.1679	0.1679	0.002630	0.001745
	Stochastic with lagged effects	0.1062	0.0882	0.1092	0.1092	0.001427	0.003213